



bag 1



bag 2

Let

E_1 = 2 white balls are transferred from bag 1 to bag 2

E_2 = 2 black balls are transferred from bag 1 to bag 2

E_3 = 1 white ball or 1 black ball are transferred from bag 1 to bag 2.

$$P(E_1) = \frac{10C_2}{13C_2} = \frac{45}{78} = \frac{15}{26}$$

$$P(E_2) = \frac{3C_2}{13C_2} = \frac{3}{78} = \frac{1}{26}$$

$$P(E_3) = \frac{10C_1 \times 3C_1}{13C_2} = \frac{30}{78} = \frac{5}{13}$$

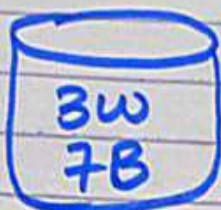
→ Let A be the event a ball (white ball) is drawn from bag 2.



→ 2 white balls

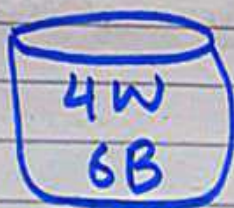
bag 2

$$P(A/E_1) = \frac{5}{10} = \frac{1}{2}$$



→ 2 black balls

$$P(A/E_2) = \frac{3}{10}$$



1 white or 1 black

$$P(A/E_3) = \frac{4}{10} = \frac{2}{5}$$

→ we have to find $P(A)$

$$P(A) = P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + P(E_3) \cdot P(A/E_3)$$

$$= \frac{15}{26} \cdot \frac{1}{2} + \frac{1}{26} \cdot \frac{3}{10} + \frac{5}{13} \cdot \frac{2}{5}$$

$$= \frac{15}{52} + \frac{3}{260} + \frac{10}{65}$$

$$= \frac{75 + 3 + 40}{260} = \frac{118}{260} = \frac{59}{130} = 0.45$$

Solutions:-

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Let's define the events

 A : The bolt is produced by machine A B : The bolt is produced by machine B. C : The bolt is produced by machine C. D : The bolt is defectiveprobabilities

$$P(A) = 0.25$$

$$P(B) = 0.35$$

$$P(C) = 0.40$$

Defected bolts

$$P(D/A) = 0.02$$

$$P(D/B) = 0.04$$

$$P(D/C) = 0.05$$

We have to find the probability that a defected bolt came from machine A.

$$P(A|D) = \frac{P(A) \cdot P(D|A)}{P(A) \cdot P(D|A) + P(B) \cdot P(D|B) + P(C) \cdot P(D|C)}$$

$$= \frac{(0.25)(0.02)}{(0.25)(0.02) + (0.35)(0.04) + (0.40)(0.05)}$$

=

$$\frac{0.005}{0.0415} = 0.0121$$

= 1.21% from machine A
the defective bolts are

✓ 1.21%

→ Continuous Random Variable

A random variable which takes infinite number of values in an interval is called continuous random variable.

Example:

→ The weight of a group of individuals.

→ Height of group of individuals

→ Price of house

→ Probability Density function (P.d.f)

A function $f(x)$ is P.d.f if

① $f(x) \geq 0$

② $\int_{-\infty}^{\infty} f(x) dx = 1$

→ Also known as density function

Example:-

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a) Find the value of k so that the function $f(x)$ defined as

$$f(x) = kx, 0 < x < 2$$

Solution:-

Must satisfy the two condition of pdf.

i. $f(x) > 0$

ii.
$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$1 = \int_{-\infty}^0 f(x) dx + \int_0^2 f(x) dx + \int_2^{\infty} f(x) dx$$

$$\int_{-\infty}^0 kx dx + \int_0^2 kx dx + \int_2^{\infty} kx dx = 1$$

$$0 + \int_0^2 kx dx + 0 = 1$$

$$k \left[\frac{x^2}{2} \right]_0^2 = 1$$

$$k \left[\frac{(2)^2}{2} - \frac{(0)^2}{2} \right] = 1$$

$$k \left[\frac{4}{2} \right] = 1$$

$$k \cdot 2 = 1$$

$k = 1/2$

b. Compute $P(X=1)$

$$P(X=1) = 0$$

c. Compute $P(X > 1)$

$$P(X > 1) = \int_1^{\infty} f(x) dx \Rightarrow k \int_1^{\infty} x dx = 1$$

$$\frac{1}{2} \int_1^2 u \, du$$

$$\frac{1}{2} \left[\frac{u^2}{2} \right]_1^2$$

$$\frac{1}{2} \left[\frac{(2)^2}{2} - \frac{(1)^2}{2} \right]$$

$$\frac{1}{2} \left[\frac{4-1}{2} \right] =$$

$$\frac{1}{2} \left[\frac{3}{2} \right]$$

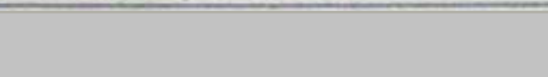
$$\frac{1}{2} \left[\frac{3}{2} \right] = \frac{3}{4}$$

d. Compute the distribution function $F(x)$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du$$

for x $-\infty < x \leq 0$

$$F(x) = \int_{-\infty}^x 0 du = 0$$



$$F(x) = \int_{-\infty}^0 0 du + \int_0^x \frac{u^2}{2} du$$

$$\left[\frac{u^3}{2+2} \right]_0^x$$

$$= \frac{x^3}{4} - \frac{(0)^3}{2} = \frac{x^3}{4}$$

Now $x > 2$

$$F(x) = \int_{-\infty}^0 0 du + \int_0^2 \frac{u}{2} du + \int_2^{\infty} 0 du = 1$$

$$= \frac{x^2}{4}$$

lec # 27
Example # 1

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$$f(x, y) = \frac{x(1+3y^2)}{4}$$

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad E(X), E(Y)$$

$$= \int_0^1 x \left(\frac{1+3y^2}{4} \right) dy$$

$$= \frac{1}{4} \int_0^1 (x + 3xy^2) dy$$

$$= \frac{1}{4} \left[xy + \frac{3xy^3}{3} \right]_0^1$$

$$= \frac{1}{4} \left[xy + xy^3 \right]_0^1$$

$$= \frac{1}{4} \left[x(1) - x(0) + x(1)^3 - x(0)^3 \right]$$

$$= \frac{1}{4} \left[x + x \right] = \frac{1}{4} [2x] = \frac{x}{2}$$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$= \int_0^2 \frac{x(1+3y^2)}{4} dx$$

$$= \frac{1}{4} \int_0^2 (x + 3xy^2) dx$$

$$= \frac{1}{4} \left[\frac{x^2}{2} + 3xy^2 \right]_0^2$$

$$= \frac{1}{4} \left[\frac{(2)^2}{2} - \frac{(0)^2}{2} + 3(1)y^2 + \underline{3(0)y^2} \right]$$

$$\frac{1}{4} \left[\cancel{2} + 3y^2 \right]$$

$$\frac{1}{2} [1 + 3y^2]$$

$$E(X+Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) f(x,y) dx dy$$

$$= \int_0^2 \int_0^1 (x+y) \frac{x(1+3y^2)}{4} dy dx$$

$$= \int_0^2 \int_0^1 (x+y) \frac{(x+3xy^2)}{4} dy dx$$

$$= \int_0^2 \int_0^1 \frac{(x^2 + 3x^2y^2 + yx + 3xy^3)}{4} dy dx$$

$$= \int_0^2 \int_0^1 \frac{x^2 + 3x^2y^2}{4} dx dy + \int_0^2 \int_0^1 \frac{xy + 3xy^3}{4} dy dx$$

$$= \int_0^2 \frac{1}{4} \left[n^2 y + n^2 y^3 \right]_0^1 dn$$

$$= \int_0^2 \frac{1}{4} \left[n^2(1) - n^2(0) + n^2(1)^3 - n^2(0)^3 \right] dn$$

$$= \int_0^2 \frac{1}{4} \left[n^2 + n^2 \right] dn$$

$$= \int_0^2 \frac{1}{2} n^2 dn$$

$$= \int_0^2 \frac{1}{2} n^2 dn$$

$$= \frac{1}{2} \int_0^2 n^2 dn$$

$$= \frac{1}{2} \left[\frac{n^3}{3} \right]_0^2 \Rightarrow \frac{1}{2} \left[\frac{(2)^3}{3} - \frac{(0)^3}{3} \right]$$

part ①

$$= \frac{1}{2} \left[\frac{8}{3} \right] \Rightarrow \frac{4}{3}$$

$$= \int_0^2 \int_0^1 \frac{xy + 3xy^3}{4} dy dx$$

$$= \int_0^2 \frac{1}{4} \left[\frac{xy^2}{2} + \frac{3xy^4}{4} \right]_0^1 dx$$

$$= \int_0^2 \frac{1}{4} \left[\frac{x(1)^2}{2} - \frac{x(0)^2}{2} + \frac{3x(1)^4}{4} - \frac{3x(0)^4}{4} \right] dx$$

$$= \int_0^2 \frac{1}{4} \left[\frac{x}{2} + \frac{3x}{4} \right] dx$$

$$= \frac{1}{4} \left[\frac{x^2}{2 \times 2} + \frac{3x^2}{2 \times 4} \right]_0^2$$

$$= \frac{1}{4} \left[\frac{x^2}{4} + \frac{3x^2}{8} \right]_0^2$$

$$= \frac{1}{4} \left[\frac{(2)^2}{4} + \frac{3(2)^2}{8} \right]$$

$$= \frac{1}{4} \left[\frac{4}{4} + \frac{12}{8} \right]$$

part 2

$$= \frac{1}{4} \left[\frac{1+3}{2} \right] = \frac{5}{8}$$

LCM

$$\frac{2+3}{2} = \frac{5}{2}$$

$$= \frac{4}{3} + \frac{5}{8} \quad (\text{By Taking LCM})$$

$$= \frac{47}{24}$$

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x,y) dx dy$$

$$= \int_0^2 \int_0^1 (xy) \cdot \frac{n(1+3y^2)}{4} dy dx$$

$$= \int_0^2 \int_0^1 \frac{xy^2 + 3xy^3}{4} dy dx$$

$$= \int_0^2 \frac{1}{4} \left[\frac{x^2 y^2}{2} + \frac{3x^2 y^4}{4} \right]_0^1 dx$$

$$= \int_0^2 \frac{1}{4} \left[\frac{x^2(1)^2}{2} - \frac{x^2(0)}{2} + \frac{3x^2(1)}{4} - \frac{3x^2(0)}{4} \right] dx$$

$$= \int_0^2 \frac{1}{4} \left[\frac{x^2}{2} + \frac{3x^2}{4} \right] dx$$

$$= \int_0^2 \frac{1}{4} \left[\frac{2x^2 + 3x^2}{4} \right] dx$$

$$\int_0^2 \frac{1}{4} \left(\frac{5x^2}{4} \right) dx$$

$$\frac{1}{4} \left[\frac{5x^3}{3 \times 4} \right]_0^2$$

$$\frac{1}{4} \left[\frac{5x^3}{12} \right]_0^2 \Rightarrow \frac{1}{4} \left[\frac{5(2)^3}{12} - \frac{5(0)^3}{12} \right]$$

$$= \frac{1}{4} \left[\begin{array}{r} 20 \\ 40 \\ 12 \\ 8 \end{array} \right]$$

$$= \frac{1}{4} \left[\begin{array}{r} 5 \\ 10 \\ 3 \end{array} \right]$$

$$= \frac{5}{6}$$

i $E(X) + E(Y) = \frac{4}{3} + \frac{5}{8}$

ii $E(X) E(Y) = \frac{47}{24}$

$$\left(\frac{4}{3} \right) \left(\frac{5}{8} \right) = \frac{20}{24} = \frac{5}{6}$$

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lecture # 25

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Example # 1

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$$f(x) = \frac{x}{2}$$

$$P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right)$$

$$= \frac{P\left(\frac{1}{3} \leq X \leq \frac{1}{2}\right)}{P\left(\frac{1}{3} \leq X \leq \frac{2}{3}\right)}$$

$$= \frac{\int_{1/3}^{1/2} \frac{x}{2} dx}{\int_{1/3}^{2/3} \frac{x}{2} dx}$$

$$= \frac{\left[\frac{x^2}{2 \times 2} \right]_{1/3}^{1/2}}{\left[\frac{x^2}{2 \times 2} \right]_{1/3}^{2/3}}$$

$$= \frac{\left[\frac{x^2}{4} \right]_{1/3}^{1/2}}{\left[\frac{x^2}{4} \right]_{1/3}^{2/3}}$$

$$= \frac{\left(\frac{1}{2}\right)^2 - \left(\frac{1}{3}\right)^2}{\frac{\left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2}$$

$$= \frac{\frac{1}{4} - \frac{1}{9}}{\frac{4}{9} - \frac{1}{9}}$$

$$= \frac{\frac{1}{4} \times 4 - \frac{1}{9} \times 4}{\frac{4}{9} \times 4 - \frac{1}{9} \times 4}$$

$$= \frac{\frac{4}{9} \times 4 - \frac{1}{9} \times 4}{\frac{4}{9} \times 4 - \frac{1}{9} \times 4}$$

$$= \frac{\frac{4}{9} - \frac{4}{9}}{\frac{16}{9} - \frac{4}{9}}$$

$$= \frac{1 - 4}{16 - 4}$$

$$= \frac{9-4}{9}$$

$$\frac{\cancel{12}^4}{\cancel{9}_3}$$

$$= \frac{5 \times 3}{9 \times 4}$$

$$= \frac{15}{36}$$

Example #02 page 185-186

A continuous random variable

X has d.f $F(x)$ as follows:

$$F(x) = 0, \text{ for } x < 0$$

$$= \frac{2x^2}{5}, \quad 0 < x \leq 1$$

$$= \frac{-3}{5} + \frac{2}{5} \left(\frac{3x - x^2}{2} \right), \quad 1 < x \leq 2;$$

$$= 1 \quad x < 2$$

find p.d.f and $P(|X| < 1.5)$

$$f(x) = \frac{d}{dx} F(x)$$

$$F(x) = 0, \quad \underline{f(x)} = 0$$

$$F(x) = \frac{2x^2}{5}$$

$$\underline{f(x)} = \frac{2(2)x}{5} = \frac{4x}{5}$$

$$F(x) = -\frac{3}{5} + \frac{2}{5} \left(3x - \frac{x^2}{2} \right)$$

$$f(x) = 0 + \frac{2}{5} \left(3 - \frac{2x}{2} \right)$$

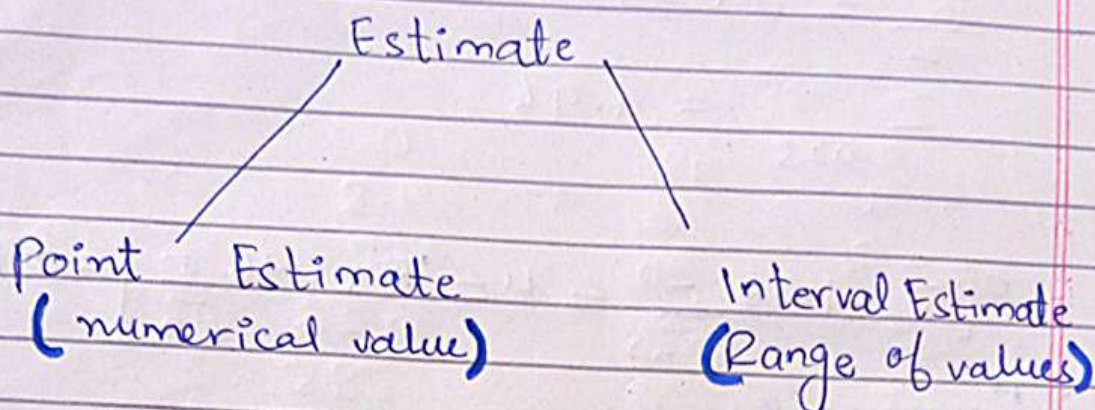
$$\underline{f(x)} = \frac{2}{5} (3 - x)$$

$$f(x) = 1$$

$$\underline{f(x)} = 0$$

Lecture #35

Confidence Interval



- Interval estimate of population parameters are called confidence interval.

Point Estimate $\bar{x} \pm 2 \left(\frac{\sigma}{\sqrt{n}} \right)$ margin of error

$\hat{p} \pm 2 \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

→ Confidence Interval for μ

95% confidence interval for μ

$$P \left[\bar{X} - Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right] = 1 - \alpha$$

$\Rightarrow$

$$1 - \alpha = 95\%$$

$$\alpha = 95\% - 1 \Rightarrow \alpha = 0.95 - 1$$

$$\boxed{\alpha = 0.05}$$

$$\frac{Z_{\alpha}}{2} = ?$$

$$\alpha = 0.05$$

$$\frac{\alpha}{2} = 0.025$$

$$Z_{0.025} = 1.96$$

$$P\left[\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}}\right] = 0.95$$

Example:

x	Number of Day	No of Emp(%)
	None	48
$\frac{1+2=1.5}{2}$	1 or 2	43
	3 or 4	90
	5 or 6	186
	7 or 8	78
	9 to 12	78 34
	13 to 20	21
	Total	500

Construct 95% confidence interval.

$$n = 500$$

if n is equal to or greater 30 we will use this formula:-

$$P\left[\bar{x} - Z_{\alpha/2} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + 1.96 \frac{s}{\sqrt{n}}\right]$$

So we have to find

$$Z_{\alpha/2} = ? \quad s = ? \quad n = 500 \quad \bar{x} = ?$$

$$1.96$$

"Confidence interval for the difference b/w the mean of the two population"

$$\mu_1 - \mu_2$$

$$\bar{X}_1 - \bar{X}_2 \pm z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$\bar{X}_1 - z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq \bar{X}_2 + z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Example

90% confidence

$$1 - \alpha = 90\%$$

$$\alpha = 90\% - 1$$

$$-\alpha = 0.9 - 1$$

$$+\alpha = +0.1$$

$$\alpha = 0.1$$

$$\frac{\alpha}{2} = 0.05$$

$$z_{\frac{\alpha}{2}} = 1.645$$

By using area Table

Everything is given just put values.

Confidence interval for a population proportion. →

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

let 92% confidence interval.

$$1 - \alpha = 92\%$$

$$-\alpha = 92\% - 1$$

$$-\alpha = 0.92 - 1$$

$$-\alpha = -0.08$$

$$\frac{\alpha}{2} = 0.04$$

$$z_{\alpha/2} = z_{0.04} = 1.75$$

$$\hat{p} = \frac{x}{n}$$

Confidence interval $p_1 - p_2$

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$\hat{p}_1 - \hat{p}_2 - z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \leq p_1 - p_2 \leq$$

$$\hat{p}_1 - \hat{p}_2 + z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Sample Size for Estimation population proportion.

$$\hat{p} = z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

Sample size for Estimation population mean

$$\underbrace{|\bar{x} - \mu|}_{\text{Statistic}} = z_{\alpha/2} \underbrace{\left(\frac{\sigma}{\sqrt{n}} \right)}_{\text{Standard error}}$$

Sampling error

↓

$$e = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

multiply by $\sqrt{n}$

$$\sqrt{n} e = z_{\alpha/2} \sigma$$

Divide the e both sides

$$\sqrt{n} \frac{e}{e} = \frac{z_{\alpha/2} \sigma}{e}$$

$$\sqrt{n} = \frac{z_{\alpha/2} \sigma}{e}$$

Square

$$n = \left(\frac{z_{\alpha/2} \sigma}{e} \right)^2$$

s = Standard deviation

$\bar{x}$ = mean.

$n = 500$

$$\bar{x} = \frac{\sum fx}{n}$$

x	f	fx
0	48	0
1.5	43	64.5
3.5	90	315
5.5	186	1023
7.5	78	585
10.5	34	357
16.5	21	346.5

$$\sum fx = 2691$$

$$\bar{x} = 5.382$$

$$s = 3.53$$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

$$\left(5.38 - 1.96 \frac{3.53}{\sqrt{500}} \leq \mu \leq 5.38 + 1.96 \frac{3.53}{\sqrt{500}} \right)$$

$$\left(5.38 - 1.96 (0.157) \leq \mu \leq 5.38 + 1.96 (0.157) \right)$$

$$5.38 - 0.309 \leq \mu \leq 5.38 + 0.309$$

$$5.071 \leq \mu \leq 5.689$$

Lecture # 37

Concept of Hypothesis Testing

Example:

A random sample of 100 workers with children in day care show mean day-care cost of Rs 2650 and a standard deviation of Rs 500. Verify the department's claim that the mean exceeds Rs 2500 at the 0.5 level with this information. α

Solution:

Step 1:

Null hyp $\rightarrow H_0 : \mu < 2500$

Alternative $\rightarrow H_1 : \mu > 2500$

Step 2:

level of significance at α
 $= 0.05$

Step 3:

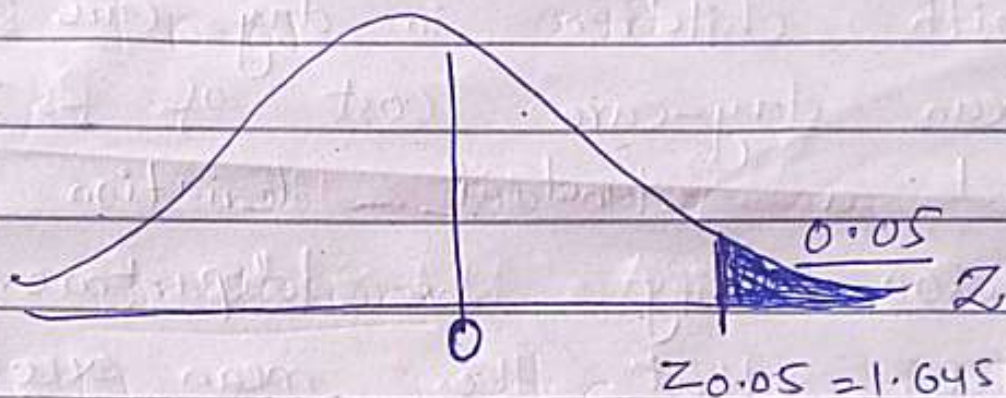
Test - Statistic

$$Z = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

Step 4

Critical region

$$Z_{0.05} = 1.645$$



Step 5:

computing the Z value

$$Z = \frac{2650 - 2500}{\frac{500}{\sqrt{100}}}$$

$$Z = 3$$

Step 6:

Conclusion.

Since the calculated value of Z is 3 then ~~and~~ 1.645

$$3 > 1.645$$

falls in rejection region. and
there fore we reject H_0
1u 38

Hypothesis - Testing $\mu_1 - \mu_2$

Example:

Step 1:

$$H_0 : \mu_1 < \mu_2$$

$$H_A : \mu_1 > \mu_2$$

Step 2:

level of significance α

$$5\% = 0.05$$

Step 3:

Test - Statistic

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Critical Region.

It's a right tailed test

$$Z_{0.05} \rightarrow Z = 1.645$$

Step 5:
Computing values.

$$= \frac{12.80 - 11.25}{\sqrt{\frac{64}{160} + \frac{47}{220}}}$$

$$= \frac{1.55}{\sqrt{0.61}} = \frac{1.55}{0.78} = 1.99.$$

Step 6:
conclusion.

1.99 is greater than 1.645.
H₀ should be rejected. so
H_A will be accepted.